

1.

$$\begin{aligned} a &\equiv a' \pmod{n} \iff \exists i \in \mathbb{Z}: a' = a + in \\ b &\equiv b' \pmod{n} \iff \exists j \in \mathbb{Z}: b' = b + jn \\ a' + b' &\equiv a + in + b + jn \equiv a + b + n(i + j) \equiv a + b \pmod{n} \end{aligned}$$

Therefore $a + b \equiv a' + b' \pmod{n}$.

$$a'b' \equiv (a + in)(b + jn) \equiv ab + ajn + inb + injn \equiv ab + n(aj + ib + injn) \equiv ab \pmod{n}$$

Therefore $ab \equiv a'b' \pmod{n}$.

2.

$$\begin{aligned} &\nexists 0^{-1}, 2^{-1}, 4^{-1}, 6^{-1} \pmod{8} \\ &1^{-1} \equiv 1 \pmod{8} \\ &3^{-1} \equiv 3 \pmod{8} \\ &5^{-1} \equiv 5 \pmod{8} \\ &7^{-1} \equiv 7 \pmod{8} \\ &\exists a^{-1} \pmod{8} \iff \gcd(a, 8) = 1 \end{aligned}$$

3.

$$\begin{aligned} n &= p_1^{a_1} \dots p_k^{a_k} \\ \text{let } a &= p_1, b = p_1^{a_1-1} \prod_{i=2}^k p_i^{a_i} \\ a &\not\equiv 0, b \not\equiv 0 \pmod{n} \\ ab &\equiv n \equiv 0 \pmod{n} \end{aligned}$$

Let's assume $\mathbb{Z}/n\mathbb{Z}$ is a field:

$$\begin{aligned} \exists a^{-1}: a^{-1}a &\equiv 1 \pmod{n} \\ a^{-1}ab &\equiv a^{-1}0 \pmod{n} \\ (a^{-1}a) &\equiv 1 \implies b \equiv 0 \pmod{n} \end{aligned}$$

which is a contradiction since we've established previously that $b \not\equiv 0 \pmod{n}$, therefore $\mathbb{Z}/n\mathbb{Z}$ is not a field.

4.

$$\begin{aligned} [a] \neq [0], p \nmid a &\implies \gcd(a, p) = 1 \implies ax + py = 1 \implies ax + py \equiv 1 \pmod{p} \implies \\ ax &\equiv 1 \pmod{p} \implies [x] \equiv [a]^{-1} \end{aligned}$$

5.

$$\begin{aligned} 1^4 &\equiv 1 \pmod{5} \\ 2^4 &\equiv 1 \pmod{5} \\ 3^4 &\equiv 1 \pmod{5} \\ 4^4 &\equiv 1 \pmod{5} \end{aligned}$$

Conj.: p is prime, $p \nmid a \implies a^{p-1} \equiv 1 \pmod{p}$.